ML Project Proposal

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1. Question

How can we apply Iterative Learning Control to successfully prepare and manipulate quantum states on hardware?

2. Motivation

2.1 Why do we care about quantum computing?

Quantum computers and simulators promise advantages over their classical counterparts, providing key insights into quantum many-body systems in material science, quantum chemistry, and high-energy physics, as well as potentially enhancing the performance in solving exponentially hard optimization problems. This long-term promise, however, relies on the development of fully fault-tolerant quantum devices.

2.2 Why do we care about quantum control?

The effectiveness of quantum devices relies heavily on the quality of prepared quantum states and their control using quantum gates. These systems are described by model parameters, such as transition frequencies, coupling strengths, and anharmonicities, as well as by constraints, such as coherence times, ionization limits, and various perturbative limits. Subject to such constraints and unitary fidelity objectives, researchers have turned to the field of quantum optimal control (QOC) for solving appropriate optical, microwave, or laser-pulse drives. While such solutions are highly effective under the simulated evolution of perfectly-defined nominal Hamiltonians [1], errors on physical hardware quickly accumulate due to miscalculated model parameters, system drift, and unknown Hamiltonian matrix elements.

2.3 Why do we care about closed-loop quantum control?

When using a static theoretical effective Hamiltonian to model a system, simulations and real-time control are vulnerable to Hamiltonian inaccuracies and hardware imperfections – this is called open-loop quantum control. Closed-loop quantum control uses iterative feedback from measurement to improve a working model based on a given theoretical Hamiltonian. Specifically, we aim to apply the classical method of Iterative Learning Control (ILC) on quantum hardware to correct nearly-optimized simulation pulses and simultaneously reduce uncertainty on system Hamiltonian parameters.

3. Method – Why do we care about Iterative Learning Control?

Current closed-loop approaches involving Reinforcement Learning (RL) use no information about the system model (i.e. the Hamiltonian) [2]. These “black-box” methods require many pulse evaluations and direct measurement of model drift (e.g. transition frequency error) as inputs to the RL network.

Iterative Learning Control acts instead as a first-order “grey-box” model. It utilizes model information to correct deviations in a system’s state during its evolution to correct the pulse at subsequent time-steps. These corrections can also be used to determine the error in the model’s parameters.

4. Data/Simulator

Our proposed platform utilizes superconducting qubits, specifically on a cavity circuit quantum electrodynamics (cavity cQED) system.

5. Project timeline

Now: We plan to use Harmoniq’s quantum toolbox to set up a simple initial problem with a nominal model, and find the optimal control pulse. One initial candidate is a two-qubit interaction such as a Hadamard gate.

By the Money Plot Check date (Oct. 30): Define a “noisy” hardware model using parameters that vary from simulation’s nominal initial model. This is to test ILC’s capability to converge towards the “noisy”

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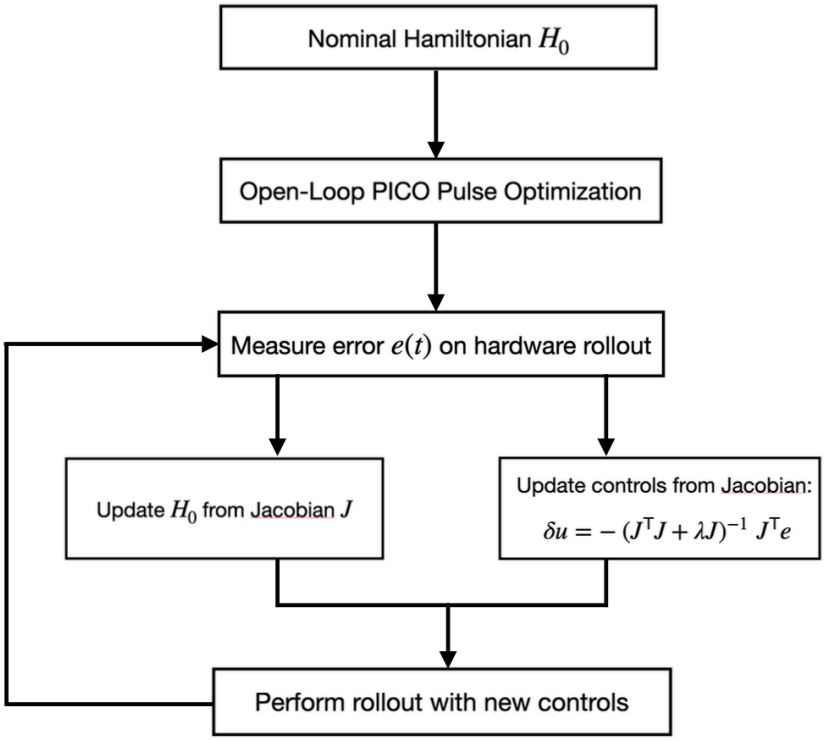


Figure 1: Optimizing quantum optimal control (QOC) using closed-loop Iterative Learning Control (ILC).

Hamiltonian parameters. We will show the method's ability to correct different Hamiltonian matrix elements in a heatmap representation of percent errors based on target and actual results. This will explore the method's ability to correct qubit frequencies, discover unmodeled interactions, and account for higher-level transitions.

By the final presentation date (Dec 16-22): We plan to have a fleshed out research report in NeurIPS format with results. We hope to explore more complex systems which require a larger set of classical observables for residual error calculation (>2 qubits, qubit-cavity systems).

Bibliography

[1] A. Trowbridge, A. Bhardwaj, K. He, D. I. Schuster, and Z. Manchester, "Direct Collocation for Quantum Optimal Control," arXiv.org. Accessed: Oct. 05, 2026. [Online]. Available: https://arxiv.org/abs/2305.03261v3

[2] A. Jaouadi and S. Ashhab, "Reinforcement Learning for Robust Calibration of Multi-Qudit Quantum Gates," arXiv.org. Accessed: Oct. 05, 2026. [Online]. Available: https://arxiv.org/abs/2604.19990v3